

Nuclear Astrophysics

Formation and Survival of Carbon-12

The Triple-Alpha Process & Helium Burning

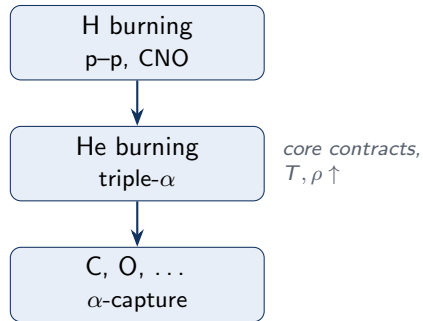
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Based on lectures by Prof. Anil Kumar Gourishetty, IIT Roorkee

July 3, 2026

Where we are in the story

- Since the Big Bang (~ 13.8 Gyr ago) nuclear reactions have driven the **synthesis of the elements**.
- Earlier: the **p-p chain** and the **CNO cycle** burn hydrogen in main-sequence stars.
- **Today's question:** once hydrogen is exhausted, what is its “ash”, and how does it build heavier nuclei?



The puzzle

Nature shows **stability gaps** at mass numbers $A = 5$ and $A = 8$. How are heavier elements built across them?

From main sequence to the super red giant

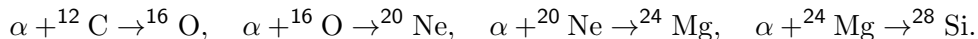
- ① Hydrogen is exhausted in the core $\Rightarrow$ the **helium core contracts**, density and temperature rise.
- ② **Simultaneously**, hydrogen ignites in a **shell** around the core.
- ③ Shell burning inflates the envelope — the radius can grow to $\sim 50\times$ its original size.
- ④ The expanding surface cools; its peak wavelength shifts into the **red** $\Rightarrow$ a **super red giant**.

Core composition

The core is now **98–99% helium**. With so many α particles available, α -capture reactions run continuously.

The naive α -capture ladder — and where it stops

If α -capture simply continued, we would expect:



- In reality quiescent helium burning essentially **stops at ${}^{16}\text{O}$** .
- There is a beautiful nuclear-structure reason — we reach it in the second half of the lecture.
- **First**, the harder question: how is ${}^{12}\text{C}$ made *at all* across the $A = 8$ gap?

Why ${}^{12}\text{C}$ matters

${}^{12}\text{C}$ is the **4th most abundant** element in the universe and the foundation of organic chemistry — and life.

Step 1 — Salpeter: $\alpha + \alpha \rightleftharpoons {}^8\text{Be}$

- Salpeter proposed building ${}^{12}\text{C}$ as a **two-step** process, needing *no* stable $A = 5$ or $A = 8$ nucleus.
- Step 1: $\alpha + \alpha \rightarrow {}^8\text{Be}$ in its ground state.
- **Problem:** ${}^8\text{Be}$ lives only $\sim 10^{-16}\text{ s}$ and decays straight back to two α 's.
- It sits just **92 keV** above the $\alpha + \alpha$ threshold.



BE/nucleon: $\alpha = 7073\text{ keV}$, ${}^8\text{Be} = 7062.44\text{ keV}$

Equilibrium

$\alpha + \alpha \rightleftharpoons {}^8\text{Be}$ reaches a tiny but real equilibrium abundance of ${}^8\text{Be}$.

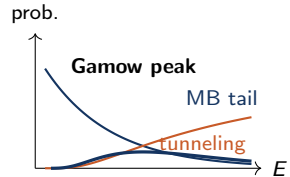
Gamow peak for $\alpha + \alpha$

Because both partners are charged, fusion proceeds through the **Gamow window**, not at the classical threshold.

Numbers to remember

- Gamow energy: $E_G \equiv b^2 \approx 31.4 \text{ MeV}$.
- Substitute into the Gamow-peak centroid E_0 .
- Fusion becomes possible for $T \gtrsim 10^8 \text{ K}$.

Different nuclei ignite in different temperature windows (10^7 – 10^{10} K). Knowing the relevant T for each reaction is part of the course.



Step 2 — the resonant capture $\alpha + {}^8\text{Be} \rightarrow {}^{12}\text{C}$

- In the burning region the α “transit time” is $\sim 10^{-19}\text{ s}$ — much shorter than ${}^8\text{Be}$ ’s 10^{-16} s life.
- So **before ${}^8\text{Be}$ decays, it can capture a third α** . This is the turning point in making ${}^{12}\text{C}$.
- The Q -value of $\alpha + {}^8\text{Be}$ is $\approx 7.366\text{ MeV}$.

The abundance problem

A *normal* (non-resonant) formation of ${}^{12}\text{C}$ badly under-predicts the observed carbon abundance.

————— 7.644 MeV 0^+ (Hoyle)
----- $\alpha + {}^8\text{Be}$ -----

————— 4.44 MeV 2^+

————— ${}^{12}\text{C}_{\text{gs}} 0^+$

Hoyle's prediction — a resonance must exist

Hoyle's insight (Nobel-worthy)

Theory and observation agree **only if** ^{12}C is produced in an excited state via a **resonance** just above the $\alpha + {}^8\text{Be}$ threshold.

- Predicted Gamow-peak centroid for $\alpha + {}^8\text{Be}$: $\sim 230\text{--}280\text{ keV}$.
- Threshold + centroid $\Rightarrow$ a ^{12}C state near

$$E_x \approx 7.366\text{ MeV} + 0.28\text{ MeV} \approx 7.65\text{ MeV}.$$

- This is the **Hoyle state** at 7.644 MeV , $J^\pi = 0^+$.
- Ground state is also 0^+ , so the direct $0^+ \rightarrow 0^+$ γ transition is **forbidden**.

Resonant reaction-rate formalism

For a charged-particle resonant reaction,

$$\langle \sigma v \rangle \propto \omega \gamma \exp\left(-\frac{E_R}{kT}\right), \quad E_R = E_x(\text{Hoyle}) = 7.644 \text{ MeV}.$$

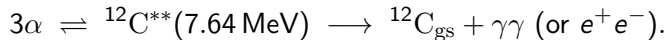
Resonance strength (partial widths):

$$\omega \gamma = \frac{\Gamma_\alpha \Gamma_{\text{rad}}}{\Gamma}, \quad \Gamma_{\text{rad}} = \Gamma_\gamma + \Gamma_{\text{pair}}, \quad \Gamma = \Gamma_\alpha + \Gamma_\gamma + \Gamma_{\text{pair}}.$$

- The Hoyle state de-excites by a γ **cascade** $7.644 \rightarrow 4.44 \rightarrow 0 \text{ MeV}$ (direct $0^+ \rightarrow 0^+$ forbidden), or by e^+e^- **pair** emission.
- Rate of the third step: $r_{3\alpha} = N_{8\text{Be}} N_\alpha \langle \sigma v \rangle_{8\text{Be}+\alpha}$.

Equilibrium, leakage, and the closed-form rate

Compact notation for the two equilibria with a small leak to the ground state:



- The electromagnetic decay is a **tiny leak** — only **~ 1 in 2500** — so equilibrium really is reached.
- Combining with the Saha equation gives the stellar triple- α rate.

Triple- α rate

$$r_{3\alpha} = \frac{N_\alpha^3}{2} 3^{3/2} \left(\frac{2\pi\hbar^2}{M_\alpha kT} \right)^3 \frac{\omega\gamma}{\hbar} e^{-Q/kT}$$

Measurables: the Q -value and the resonance strength $\omega\gamma$.

Temperature sensitivity

Energy generation scales as $\varepsilon_{3\alpha} \propto T^{41}$ — extraordinarily sensitive to temperature.

Why you cannot do $\alpha + {}^8\text{Be}$ directly

Two showstoppers

- ① ${}^8\text{Be}$ lives only $\sim 10^{-16}\text{ s} \Rightarrow$ **no ${}^8\text{Be}$ target** can be made.
- ② The relevant transition is $0^+ \rightarrow 0^+$, i.e. **γ -forbidden**.

Solution: use **indirect** measurements.

Indirect routes

- **Inelastic scattering:** ${}^{12}\text{C}(p, p'){}^{12}\text{C}^*$ populates the Hoyle state, which breaks up into 3α .
- Detect the three α 's in **coincidence** (Si surface-barrier or CsI detectors) $\rightarrow$ cross section.
- **${}^{12}\text{B}$ implantation:** implant ${}^{12}\text{B}$ in a solid-state counter; its breakup yields 3α .

The C/O ratio sets the stage

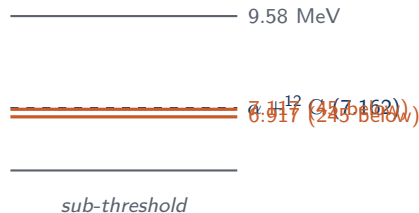
Observed fact

The carbon-to-oxygen ratio is $C/O \approx 0.6 < 1$. That is why oxygen is the 3rd most abundant element, just after H and He.

- Most carbon is the *direct* product of triple- α .
- For the ratio to hold, the destruction reaction $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ must proceed at a moderate rate — not too fast (else carbon is wiped out), not too slow.
- **Question:** is there a convenient resonance to speed it up, as there was for ^{12}C ?

Level scheme of ^{16}O — no helpful resonance

- $\alpha + {}^{12}\text{C}$ threshold sits at **7162 keV** in ^{16}O .
- Nearby states 7117 and 6917 keV lie **below** the threshold $\Rightarrow$ **sub-threshold resonances**.
- No resonance sits above threshold (even up to 2 GK) to accelerate the reaction.
- Gamow-peak centroid ≈ 1.46 MeV (at relevant T).



So how is O made?

Through the **tails** of the sub-threshold resonances and **non-resonant direct capture** — slow, but enough.

Why $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ is so hard to measure

- Estimated cross section in the stellar window: $\sim 10^{-17}$ b — effectively **impossible** to measure directly.
- **Background problem:** real targets contain ^{13}C ; $^{13}\text{C}(\alpha, n)^{16}\text{O}$ makes neutrons whose γ -background swamps the signal.

Two experimental work-arounds

- ① **Swap beam and target:** use a helium gas target and a ^{12}C *beam* (inverse kinematics).
- ② Use **highly enriched ^{12}C** to suppress the ^{13}C contamination.

Why the ladder halts at ^{16}O : the ^{20}Ne block

Ideally $\alpha + ^{16}\text{O} \rightarrow ^{20}\text{Ne}$ would continue the chain. It does not run efficiently:

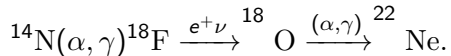
- Compound nucleus ^{20}Ne de-excites by γ ; the Q -value is low, $\approx 4.73\text{ MeV}$.
- There is a state at 4.97 MeV — but its **unnatural parity 2^-** forbids the resonant channel.
- Remaining levels are **too weak / too narrow** to help.
- Only **direct capture** ($\sigma < \text{nanobarn}$) remains — unmeasured; stellar networks rely on theory.

Result

Parity and isospin selection rules in ^{20}Ne **block quiescent helium burning** — which is why burning piles up at C and O. (At $T \gtrsim 0.3\text{ GK}$ in massive stars, $\alpha + ^{16}\text{O}$ finally overtakes $\alpha + ^{12}\text{C}$.)

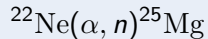
Side reactions: ^{14}N and the road to neutron sources

CNO leaves $\sim 1\text{--}2\%$ ^{14}N in the helium zone. It seeds a chain:



- ^{18}O fate: $^{18}\text{O}(\alpha, \gamma)^{22}\text{Ne}$ ($Q = 9.67$ MeV) vs. $^{18}\text{O}(\alpha, n)^{21}\text{Ne}$ ($Q = -0.7$ MeV).
- Low $T \Rightarrow$ capture dominates; high $T \Rightarrow$ neutron production dominates.

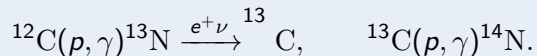
Key *s-process* neutron sources



These feed neutron capture building elements *beyond iron*.

Neutron poisons and the ^{13}C supply

Where does ^{13}C come from?



- ^{14}N can also **absorb** neutrons: $^{14}\text{N}(n, p)^{14}\text{C}$ ($Q = 0.63$ MeV).
- ^{14}C then β -decays back to ^{14}N , which can absorb again — a **neutron poison** cycle.
- **Lesson:** before crowning any reaction a neutron *source*, check whether it is also a strong neutron *sink*.

Two fine-tunings that make our universe possible

1. The tiny ${}^8\text{Be}$ mass gap

${}^8\text{Be}$ lies only ~ 922 keV above 2α . If this gap were large, the Hoyle resonance probability would collapse and ${}^{12}\text{C}$ would not form.

2. The Hoyle resonance

The 0^+ state at 7.644 MeV, just above $\alpha + {}^8\text{Be}$, is exactly what lets 3α build ${}^{12}\text{C}$ efficiently.

Three pillars of helium burning

- ① Small mass difference between 2α and ${}^8\text{Be}$.
- ② Existence of the Hoyle resonance in ${}^{12}\text{C}$.
- ③ Selection rules in ${}^{20}\text{Ne}$ that **block** further quiescent burning, holding burning at C and O.

Summary — formation & survival of ^{12}C

- **Stability gaps** at $A = 5, 8$ are bridged by the **triple- α** process inside (super) red giants.
- **Salpeter:** $\alpha + \alpha \rightleftharpoons {}^8\text{Be}$; **Hoyle:** resonant $\alpha + {}^8\text{Be} \rightarrow {}^{12}\text{C}^{**} \rightarrow {}^{12}\text{C}_{\text{gs}}$.
- Rate $\propto e^{-Q/kT}$, $\varepsilon \propto T^{41}$; measured *indirectly* (inelastic scattering, ^{12}B breakup).
- ^{12}C **survives** because $^{12}\text{C}(\alpha, \gamma)^{16}\text{O}$ is only **moderate** (no helpful resonance) $\Rightarrow \text{C/O} \approx 0.6$.
- Further burning is **blocked** at ^{16}O by ^{20}Ne selection rules; side reactions seed **s-process** neutron sources.

Next: neutron-capture nucleosynthesis — the s- and r-processes.

Thank you

Questions?